THE COMPARISON BETWEEN THE DERIVED TENSOR PRODUCT WITH THE POWERS OF AN IDEAL AND THE ORDINARY PRODUCTS IS A PRO-ISOMORPHISM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 29676-29685 (canonical tag 0G3N) and the counted proof is the proof environment at source lines 29687-29734, which immediately follows it and is its own proof.

Editorial note. The proof contains one sentence that is not terminated by a period before the next display begins, at source line 29722; the printed text is copied exactly as it stands and is not corrected.

Editorial note. The proof refers to lemma-sequence-powers-pro-bounded and lemma-consequence-Artin-Rees-bis of this chapter and to a pro-isomorphism criterion of another chapter; all three are carried as navigation stubs with their roles recorded in the reasons.

Editorial note. The chapter file pinned-original-source.tex carries only an input of the shared preamble on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus the native theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

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Allowed standard prerequisites (editorial).

- The inverse system of powers I^n is pro-isomorphic to the system generated by the n-th powers of a finite set of generators, and the auxiliary systems attached to the generators carry the same pro-information. used at proof lines 29688-29698. Replaces the original system by the system generated by the chosen generators and reduces the statement to that system.
- An isomorphism of pro-objects of a derived category can be checked on cohomology in each degree, and it suffices to compute the derived tensor product with a bounded above complex of finite free modules in place of the bounded complex M. used at proof lines 29699-29725. Reduces the pro-isomorphism of complexes to a pro-isomorphism of modules in each cohomological degree, for a convenient representative of M.
- The Artin-Rees consequence supplies the pro-isomorphism between the cohomology of the tensor product with the powers of the generators and the

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cohomology of the ordinary products. invoked at proof lines 29731-29733. Provides the final pro-isomorphism, thereby proving the lemma.

Source reference. Original lemma 0G3M: sources/stacks-more-algebra/pinned-original-source.tex, source lines 29648–29648. The reference supplies the pro-isomorphism used at proof lines 29691-29692 to replace the system of powers by the system generated by f_1, ..., f_r; the reductions that use it are inside the counted span.

Source reference. Original lemma 0G3D: sources/stacks-chow/tags.txt, source lines 20088–20088. The reference is the criterion that lets an isomorphism of pro-objects be checked on cohomology in each degree, invoked at proof lines 29709-29710; the degree bound that makes the check sufficient is established inside the counted span.

Source reference. Original lemma 0G3K: sources/stacks-more-algebra/pinned-original-source.tex, source lines 29432–29432. The reference is the Artin-Rees consequence applied at proof lines 29731-29733 to the last comparison of cohomology; the three reductions that bring the lemma into the form where this consequence applies are inside the counted span.

Lemma 0.2. *Let A be a Noetherian ring. Let $I \subset A$ be an ideal. Let $M^\bullet$ be a bounded complex of finite A-modules. The inverse system of maps*

$$I^n \otimes^{\mathbf{L}}_A M^\bullet \longrightarrow I^n M^\bullet$$

defines an isomorphism of pro-objects of $D(A)$.

Proof. Choose generators $f_1, \ldots, f_r \in I$ of I. The inverse system I^n is pro-isomorphic to the inverse system $(f_1^n, \ldots, f_r^n)$ in the category of A-modules. With notation as in Lemma 0G3M we find that it suffices to prove the inverse system of maps

$$I_n^\bullet \otimes^{\mathbf{L}}_A M^\bullet \longrightarrow (f_1^n, \ldots, f_r^n)M^\bullet$$

defines an isomorphism of pro-objects of $D(A)$. Say we have $a \leq b$ such that $M^i = 0$ if $i \notin [a, b]$. Then source and target of the arrows above have cohomology only in degrees $[-r + a, b]$. Thus it suffices to show that for any $p \in \mathbf{Z}$ the inverse system of maps

$$H^p(I_n^\bullet \otimes^{\mathbf{L}}_A M^\bullet) \longrightarrow H^p((f_1^n, \ldots, f_r^n)M^\bullet)$$

defines an isomorphism of pro-objects of A-modules, see Derived Categories, Lemma 0G3D. Using the pro-isomorphism between $I_n^\bullet \otimes^{\mathbf{L}}_A M^\bullet$ and $I^n \otimes^{\mathbf{L}}_A M^\bullet$ and the pro-isomorphism between $(f_1^n, \ldots, f_r^n)M^\bullet$ and $I^n M^\bullet$ this is equivalent to showing that the inverse system of maps

$$H^p(I^n \otimes^{\mathbf{L}}_A M^\bullet) \longrightarrow H^p(I^n M^\bullet)$$

defines an isomorphism of pro-objects of A-modules Choose a bounded above complex of finite free A-modules $P^\bullet$ and a quasi-isomorphism $P^\bullet \to M^\bullet$. Then it suffices to show that the inverse system of maps

$$H^p(I^n P^\bullet) \longrightarrow H^p(I^n M^\bullet)$$

is a pro-isomorphism. This follows from Lemma 0G3K as $H^p(P^\bullet) = H^p(M^\bullet)$. $\square$